
Project Goals & Development Process

Robotic Intervention for Pollution and Plastic Litter Extraction
(RIPPLE)

MECHTRON 4TB6: Mechatronics Capstone Design Process

Dr. Alan Wassyng

Group 14

September 30th, 2025

Kaelyn Bamford	400299080
Vedashri Pandya	400314055
Shayda Rahmatyan	400322952
Thuvarakaa Sathiyamether	400318002
Clara Yaromich	400311710
Avery Zeiler	400305001

Table of Contents

Project Description.....	1
Project Goals.....	2
Roles and Responsibilities	4
Process Workflow	6
Handling Changes	9
Version Control	11
Tools & Technology	12
Software Technology	12
Hardware Technology.....	13
Citations	14

Table of Figures

Table 1: First Level Project Goals	2
Table 2: Project Stretch Goals	3
Table 4: Project Technical Roles	5
Table 5: Project Non-Technical Roles.....	5
Table 6: Project Workflow Plan	6
Table 3: Version Control Tools	11
Table 7: Software Tools and Use Cases	12
Table 8: Hardware Tools and Uses	13

Project Description

Providing over forty million people with clean drinking water, the home of over 3500 plant and animal species, and welcoming millions of recreationalists every year to swim in their waters, the Great Lakes are a vital part of the Canadian economy and ecosystem [1]. The increasing levels of microplastics being found in animals, environments, and humans is of growing concern to health experts, with Canadian waterbodies being no exception. Over the past few decades, the Great Lakes have been facing significant challenges with surface waste accumulation, with the Rochester Institute of Technology estimating approximately 22 million pounds of pollution ending up in the Great Lakes every year [2]. If these plastics are not extracted from the water, their decomposition results in increased microplastics in drinking water, and an increased number of beach closures due to health safety, making the timely extraction of waste imperative to the continuing health of Canadian waterways.

The proposed project is to design and manufacture a scaled aquatic system (referred to as the *system*) that can maneuver on the surface of water, collecting any floating trash. The system will be remotely controllable and communicate its location back to the user. It will identify obstacles in its path and relay this information back to the user. The system will collect the trash using a Trash Collection Mechanism (TCM) and alert the user when the TCM is full.

The timeline for the project is eight months, starting in September 2025 and finishing in April 2026. Major design milestones include Proof of Concept (nine weeks into development), Revision 0 (sixteen weeks into development), and Revision 1 (twenty-seven weeks into development). The budget for this project is \$750 and will encapsulate all building materials, purchased tools, and testing equipment required for the project's completion.

Project Goals

Table 1: First Level Project Goals

First Level Goals	Description
Collect Waste on Bodies of Water	The system will allow for routine sweeping of vast areas not otherwise easily traversable, while collecting a variety of pollutants including small debris that would be difficult or time consuming to collect using other methods.
Remotely Controllable & Communicative	To increase operational flexibility in complex, dynamic aquatic environments, the system will allow for a user not in visual contact with the machine to accurately and effectively control the TCM. This includes appropriate communication of vital operational information such as the TCM's location, the state of the machine, and information about the TCM's immediate environment.
User Interface (UI)	There will be a UI which allows the user to view details from the system and interact with it.
Environmental Awareness	The system will be operated in a large variety of aquatic environments including waterways that have other moving motor bodies. The TCM will allow for a remote user to safely navigate the machine by recognizing obstacles in its path.
Capacity Detection	The system will be able to accurately identify when the TCM is full and relay this information back to the user. This feature will maximize the periods of operation between maintenance while minimizing the risk of operational failure and allow the user to reliably plan collecting routes.
Waterproof	The system's electrical components will be fully enclosed in a waterproof case to limit the impact of water exposure. This will improve both the reliability and lifetime of the TCM, limiting the chances of failure due to damaged electrical components while increasing its range of operational environments. The TCM will be able to work in choppy waters regardless of the associated splash risks.

Project Goals & Development Process

Table 2: Project Stretch Goals

Stretch Goals	Description
Controlling Trash Mechanism	User may be able to turn the TCM on and off. This allows the machine to be used in a larger variety of situations and contexts by allowing the machine to travel near objects that the user would not want removed from the environment.
Object Identification & Classification	System may be able to identify nearby objects without user input and be able to differentiate trash, obstacles, and naturally occurring aquatic plant/animals, and from this classification, know whether it should collect said object. This will minimize the impact that the TCM has on the environment and maximize the available space in the TCM being used for actual trash, while avoiding objects that could damage the machine.
Trash Hotspot Identification	Waterways oftentimes will have locations that are much more prone to the buildup of trash due to population distribution and related water currents. The TCM may be able to record common collection locations and build up a memory library of pollutant hotspots. This could reduce time spent on water not collecting and improve the overall effectiveness of the system.

Roles and Responsibilities

Each team member will be assigned the technical and non-technical roles shown in Table 4 and Table 5. Each sub-team will develop the part of the project that they are assigned, and weekly meetings will be designated for updating other sub-teams and discussing design decisions and changes. The sub-teams will receive feedback and suggestions from all team members but ultimately are responsible for making the final call for their assigned work. All team members are responsible for documenting their individual and sub-team work with sufficient detail and clarity. When disagreements arise, minor disagreements will be resolved between sub-team members, and it is expected that major disagreements which impact the work of other sub-teams will be brought to the full group for further discussion. Decisions will be made by majority vote or consensus when appropriate. Documentation will be presented during weekly meetings to inform the other team members of each sub-team's updates.

Project Goals & Development Process

Table 3: Project Technical Roles

Roles	Responsibilities	Team Member(s)
Mechanical	Design all required CAD components and select and purchase TCM and mechanical components.	Clara and Shayda
Electrical	Research, select, and purchase all electrical components and manage necessary firmware.	Kaelyn and Avery
Software	Develop the UI and assist with the firmware development.	Thuvarakaa and Vedashri

Table 4: Project Non-Technical Roles

Roles	Responsibilities	Team Member(s)
Project Manager	Create and update project deadlines through Notion.	Avery
Project Coordinator	Schedule meetings and book study rooms.	Thuvarakaa and Kaelyn
Project Administrator	Track the project's budget and submit deliverables.	Clara and Vedashri
Communication Specialist	Manage communication between the team and professors, supervisors, and external stakeholders. Create and facilitate group chats.	Shayda
Note Taker	Take meeting minutes and record important discussion points.	Rotating

Process Workflow

During the early stages of the project (generating ideas, researching, and drafting the proof of concept), all members will work together as a team with separate responsibilities delegated as needed. Once Revision 0 begins, the team will be divided into three sub-teams of two members each, namely Mechanical, Electrical, and Software as outlined in the roles and responsibilities.

Table 6 describes the steps that need to be taken to succeed in this project.

Table 5: Project Workflow Plan

#	Step/Input	Output & Acceptance Criteria	Course Deliverable	Timeline
1	Proof of Concept (POC)			
1.1	Generating Ideas	List of relevant design ideas that solve the problem.		September
1.2	Researching	List of existing design solutions for aquatic robots and TCM designs.	Project Goals & Development Process	
1.3	Drafting POC	Detailed sketches, CAD models, and workflows for the mechanical, electrical, and software aspects.	Proof of Concept with Draft Demo Contract	October
2	Revision 0			
2.1	Investigating System Requirements & Hazards	List of all necessary system requirements for mechanical, electrical, and software, and hazard analysis of potential solution.	Rev 0 – System Requirements & Hazard Analysis	November
2.2	Drafting Revision 0	Improvements made to POC sketches, CAD models, and workflows.	Rev 0 – System Design	
2.3	Purchasing	Purchase of all parts and 3D printing of any necessary components.		

Project Goals & Development Process

2.4	Building Revision 0	Physical working prototype with all mechanical, electrical and software components integrated.		December
2.5	Testing Revision 0	Accurate test cases created to effectively demo the product. Complete testing in body of water and identified problems and improvements to be made to Rev 0.	Presentation of Rev 0	January
2.6	Validating & Verifying Revision 0	All V&V tasks for the system have been developed and completed, and pass/fail criteria, along with next steps to troubleshoot or enhance design, are updated.	Rev 0 – V & V	February
3	Revision 1			
3.1	Making Changes to Revision 0	Document changes needed from Revision 0 and generate Scope of Work for Revision 1, including stretch goals that could be incorporated.		February
3.2	Designing Revision 1	Design additional components and features needed for Revision 1.		
3.3	Re-Purchasing	Draft BOM and repurchase any additional components for the building of Revision 1 (if needed).		
3.4	Building Revision 1	Working prototype of entire system consisting of mechanical, electrical and software components.	Presentation of Rev 1	March
3.5	Documenting Revision 1	All steps of design process have been clearly and thoroughly documented in an acceptable format.	Rev 1 – Documentation	

Project Goals & Development Process

3.6	Testing Revision 1	Test of Revision 1 in multiple environments to validate and verify the product. All requirements are met, and all system components have been successfully integrated.		
3.7	Presenting	Presentation has been finalized, and product has been demonstrated.	EXPO Video/Poster Presentation Final Presentation	April

Handling Changes

The team will use Notion to communicate and track hardware and design changes, while GitHub will serve as the primary tool for software version control and tracking. In addition, the team plans to further explore other version control solutions to ensure efficiency across the project. To manage progress, an Issue Tracking File will be created for objective tracking, which will include clearly defined column headers to organize and monitor tasks effectively.

To ensure that all development non-software artefact issues/changes are properly maintained, tracked, and verified, our team implemented a structured change and issue management process. This is being done through the Notion application with a database titled *Issue Tracking File*. Notion allows for the submission, disposition, assignment, and verification of issues and change requests. It allows for linking of affected artefacts, such as design files or documentation. It also provides a space to refer to test cases used and review methods for the verification process in the notes and comments section.

Team members log new issues into the Notion Issue Tracking File database using a standardized template that captures *Issue ID, Issue Title, Date Recorded, Recorder, Priority Level, Affected Artefacts, Description of Change, Resolution Assignment, Status and Type of Issue/Change (Bug, Enhancement, New Feature, Documentation)*. The status field will be set to “Not started” until team disposition begins.

Each issue undergoes a disposition process facilitated through the comments section of the relevant issue on Notion. When the new issue is submitted, the team will discuss its priority, propose solutions, and outline verification strategies in the notes section of the relevant issue. Verification can include functional testing, hardware trials, or peer review. After successful verification, the status field is changed from *In Progress* to *Resolved*, completing the lifecycle of the issue.

Project Goals & Development Process

This structured issue tracking process ensures that all changes, bug reports, enhancements, and documentation updates are systematically managed, assigned, and verified. By maintaining all discussions, verification steps and resolutions within Notion, the team achieves full traceability from issue submission to closure.

For software artefacts, our team will use GitHub for version control. GitHub provides a complete history of changes. Supports parallel development through branches and integrates with an issue tracker to ensure traceability from problem identification to resolution.

Each new software bug, enhancement, or feature request is logged as a GitHub Issue. Issues include a description, priority, and any affected artefacts. The team members tackling the issue of a change request must create a dedicated branch named after the issue (e.g., issue-23-fix-sensor) to isolate their changes. Every commit on this dedicated branch directly references the issue in the commit request and the pull request description.

When development is complete, the team member opens a Pull Request to merge their branch into the main branch. The pull request will be linked to the corresponding issue using GitHub's supported keywords such as "Fixes #23" or "Closes #23". This will automatically create a link between the pull request and the issue, and ensures the issue is closed once the pull request is merged. If we do not desire to close the request, the description will be referenced as "Related to #23". Pull requests will be reviewed by the team once the pull request is opened. Reviewing team members can examine differences between the new branch that addresses the issue and the main branch. Once a reviewer approves and all concerns are resolved, the pull request may be merged into the main branch.

Version Control

Microsoft Teams and SharePoint will be used to store and track CAD files and deliverables. When working on CAD files, the team member will upload the latest files to the *CAD* folder in the SharePoint. When another team member wants to amend that CAD file, they download them to their device and then make the changes. After they are finished, they will upload the file, renaming it to a higher revision number. Within the *CAD* folder, there will be multiple subfolders for the various parts of the product, such as *Enclosure* and *TCM*. The Notion will be used to keep track of the different revisions of the CAD files and who changed them. For software development, the team will use a GitHub repository to manage version control and change requests. Naming conventions for all CAD and software files will be consistent. All team members will make sure they follow the folder organization structure that everyone agrees to.

Table 6: Version Control Tools

Platform	Purpose
Notion	Assigning tasks for hardware and design changes, tracking progress and deadlines for tasks and deliverables.
GitHub	Managing software changes.
Microsoft SharePoint	Managing hardware changes.
Microsoft Teams	Keeping track of course deliverables.

Tools & Technology

This project will implement a range of software and hardware technologies to achieve our goal of collecting surface water trash. As detailed below, a specific use case is defined for each technology, demonstrating its relevance to our project.

Software Technology

Table 7: Software Tools and Use Cases

Software Technology	Version/ Standards	Use Case
GitHub	3.17.5	Version control for source code.
Visual Studio Code	1.104.1	Code development interface.
Python	PEP 8	All code for the UI will be written in Python.
C++	Embedded C Coding Standard	All code on the microcontroller will be written in C++.
Autodesk Inventor	2025	All 3D modeling and FEA analysis will be done.
Prusa Slicer	2.9.2	3D printing software that will be used for our mounting structures.
MATLAB/Simulink	2022b	Simulation of our design in water.

Hardware Technology

Table 8: Hardware Tools and Uses

Hardware Technology	Use Case
3D Printers	Printing of enclosures and needed additional assembly components.
Power Source	Required to power the TCM and microcontroller.
Machining/Hand Tools	For assembly of the system.
User Interface	TV display from camera for visualization.
Camera	Real time visualizer.
Microcontroller	Board to communicate with electronics.
Electronic Components: Sensors/Actuators/Wiring	Required for movement awareness and power distribution.

Citations

[1] “About the Lakes,” Great Lakes Commission, <https://www.glc.org/lakes/> (accessed Sep. 28, 2025).

[2] “Great Lakes Plastic Pollution,” Alliance for the Great Lakes, <https://greatlakes.org/issues/great-lakes-plastic-pollution-fighting-for-plastic-free-water/> (accessed Sep. 28, 2025).